

**6.2** Solution: (a) Let  $\mathbf{x}'_0$  be a solution to the vector equation

$$\mathbf{f}(\mathbf{x}') = \mathbf{x}' - \mathbf{r} \left( t - \frac{|\mathbf{x} - \mathbf{x}'|}{c} \right) = 0,$$

then the Dirac delta function  $\delta(\mathbf{f}(\mathbf{x}'))$  can be expressed as

$$\delta(\mathbf{f}(\mathbf{x}')) = \frac{\delta(\mathbf{x}' - \mathbf{x}'_0)}{|\det[J\mathbf{f}(\mathbf{x}'_0)]|},$$

where  $J\mathbf{f}(\mathbf{x}'_0)$  is the Jacobian of  $\mathbf{f}$  at  $\mathbf{x}' = \mathbf{x}'_0$ . It is straightforward to show that

$$J\mathbf{f}(\mathbf{x}'_0) = \begin{pmatrix} 1 - v_x n_x/c & -v_x n_y/c & -v_x n_z/c \\ -v_y n_x/c & 1 - v_y n_y/c & -v_y n_z/c \\ -v_z n_x/c & -v_z n_y/c & 1 - v_z n_z/c \end{pmatrix},$$

where

$$\mathbf{n} = \frac{\mathbf{x} - \mathbf{x}'_0}{|\mathbf{x} - \mathbf{x}'_0|},$$

and  $\mathbf{v}$  is the velocity of the particle at the retarded time  $t' = t - |\mathbf{x} - \mathbf{x}'_0|/c$ .

The determinant of the Jacobian can be easily calculated, with

$$\det[J\mathbf{f}(\mathbf{x}'_0)] = 1 - \frac{1}{c} (v_x n_x + v_y n_y + v_z n_z) = 1 - \frac{\mathbf{v} \cdot \mathbf{n}}{c}.$$

In fact, there is a very general result, stating that

$$\det(\delta_{ij} - a_i b_j) = 1 - \sum_i a_i b_i.$$

Let  $\kappa = 1 - \mathbf{v} \cdot \mathbf{n}/c$ , then

$$\int \delta \left[ \mathbf{x}' - \mathbf{r} \left( t - \frac{|\mathbf{x} - \mathbf{x}'|}{c} \right) \right] d^3 x' = \int \frac{\delta(\mathbf{x}' - \mathbf{x}'_0)}{|\det[J\mathbf{f}(\mathbf{x}'_0)]|} d^3 x' = \frac{1}{\kappa}.$$

(b) From Eq. (6.55), the electric field is

$$\mathbf{E}(\mathbf{x}, t) = \frac{1}{4\pi\epsilon_0} \int \left\{ \frac{\hat{\mathbf{R}}}{R^2} [\rho(\mathbf{x}', t')]_{\text{ret}} + \frac{\hat{\mathbf{R}}}{cR} \left[ \frac{\partial \rho(\mathbf{x}', t')}{\partial t'} \right]_{\text{ret}} - \frac{1}{c^2 R} \left[ \frac{\partial \mathbf{J}(\mathbf{x}', t')}{\partial t'} \right]_{\text{ret}} \right\} d^3 x',$$

where  $\hat{\mathbf{R}}$  is  $\mathbf{n}$  from part (a) and  $R = |\mathbf{x} - \mathbf{x}'_0|$ . As pointed out by Eq. (6.57),

$$\left[ \frac{\partial f(x', t')}{\partial t'} \right]_{\text{ret}} = \frac{\partial}{\partial t} [f(x', t')]_{\text{ret}},$$

and also notice that  $R$  does not depend on  $t$ , we can write the expression for electric field as

$$\mathbf{E}(\mathbf{x}, t) = \frac{1}{4\pi\epsilon_0} \int \left\{ \left[ \frac{\hat{\mathbf{R}}}{R^2} \rho(\mathbf{x}', t') \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R} \rho(\mathbf{x}', t') \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{J}(\mathbf{x}', t')}{R} \right]_{\text{ret}} \right\} d^3 x'.$$

Since  $\rho(\mathbf{x}', t') = q\delta[\mathbf{x}' - \mathbf{r}(t_{\text{ret}})]$  and  $\mathbf{J}(\mathbf{x}', t') = \rho(\mathbf{x}', t')\mathbf{v}(t')$ , following the result from part (a), we can directly write down the electric field as

$$\mathbf{E}(\mathbf{x}, t) = \frac{q}{4\pi\epsilon_0} \left\{ \left[ \frac{\hat{\mathbf{R}}}{\kappa R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}} \right\}.$$

Similarly, for the magnetic field,

$$\begin{aligned} \mathbf{B}(\mathbf{x}, t) &= \frac{\mu_0}{4\pi} \int \left\{ \left[ \rho \mathbf{v} \times \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[ \rho \mathbf{v} \times \frac{\hat{\mathbf{R}}}{R} \right]_{\text{ret}} \right\} \\ &= \frac{\mu_0 q}{4\pi} \left\{ \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} \right\}. \end{aligned}$$

(c) Let us first establish a few useful relations. Denote  $\mathbf{R} = \mathbf{x} - \mathbf{x}'(t')$ , then

$$\frac{\partial}{\partial t'} R = -\mathbf{v}(t') \cdot \hat{\mathbf{R}},$$

with

$$\mathbf{v} = \frac{d\mathbf{x}'(t')}{dt'}, \quad \hat{\mathbf{R}} = \frac{\mathbf{x} - \mathbf{x}'(t')}{|\mathbf{x} - \mathbf{x}'(t')|}.$$

Applying the chain rule, we have

$$\frac{\partial}{\partial t'} \left( \frac{1}{R} \right) = \frac{\mathbf{v}(t') \cdot \hat{\mathbf{R}}}{R^2},$$

and

$$\frac{\partial}{\partial t'} \hat{\mathbf{R}} = \frac{(\mathbf{v}(t') \cdot \hat{\mathbf{R}})\hat{\mathbf{R}} - \mathbf{v}(t')}{R}.$$

Also, for the retarded time,

$$t' = t - \frac{|\mathbf{x} - \mathbf{x}'(t')|}{c},$$

we have

$$\frac{dt}{dt'} = 1 - \frac{\mathbf{v}(t') \cdot \hat{\mathbf{R}}}{c} = \kappa.$$

Now, consider the Jefimenko expression for the magnetic field. The second term can be written as

$$\frac{\partial}{\partial t} \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} = \frac{1}{[R]_{\text{ret}}} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa} \right]_{\text{ret}} + \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa} \right]_{\text{ret}} \frac{\partial}{\partial t} \left[ \frac{1}{R} \right]_{\text{ret}}.$$

Since

$$\frac{\partial}{\partial t} \left[ \frac{1}{R} \right]_{\text{ret}} = \frac{\partial}{\partial t'} \left[ \frac{1}{R} \right]_{\text{ret}} \frac{dt'}{dt} = \frac{1}{\kappa} \frac{\partial}{\partial t'} \left( \frac{1}{R(t')} \right) = \frac{\mathbf{v}(t') \cdot \hat{\mathbf{R}}}{\kappa R(t')^2} = \left[ \frac{\mathbf{v} \cdot \hat{\mathbf{R}}}{\kappa R^2} \right]_{\text{ret}},$$

then

$$\mathbf{B}(\mathbf{x}, t) = \frac{\mu_0 q}{4\pi} \left\{ \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa R^2} \right]_{\text{ret}} + \frac{1}{c[R]_{\text{ret}}} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa} \right]_{\text{ret}} + \frac{1}{c} \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa} \frac{\mathbf{v} \cdot \hat{\mathbf{R}}}{\kappa R^2} \right]_{\text{ret}} \right\}$$

$$\begin{aligned}
&= \frac{\mu_0 q}{4\pi} \left\{ \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa R^2} \right]_{\text{ret}} \left[ 1 + \frac{\mathbf{v} \cdot \hat{\mathbf{R}}}{\kappa c} \right]_{\text{ret}} + \frac{1}{c[R]_{\text{ret}}} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa} \right]_{\text{ret}} \right\} \\
&= \frac{\mu_0 q}{4\pi} \left\{ \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa^2 R^2} \right]_{\text{ret}} + \frac{1}{c[R]_{\text{ret}}} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v} \times \hat{\mathbf{R}}}{\kappa} \right]_{\text{ret}} \right\},
\end{aligned}$$

where have used

$$1 + \frac{\mathbf{v} \cdot \hat{\mathbf{R}}}{\kappa c} = \frac{1}{\kappa}.$$

To verify the equivalence for the electric field, let us start with the Feynman expression. Notice that

$$\begin{aligned}
\frac{1}{c^2} \frac{\partial^2}{\partial t^2} [\hat{\mathbf{R}}]_{\text{ret}} &= \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{1}{\kappa} \frac{\partial \hat{\mathbf{R}}}{\partial t'} \right]_{\text{ret}} \\
&= \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{(\mathbf{v} \cdot \hat{\mathbf{R}}) \hat{\mathbf{R}} - \mathbf{v}}{\kappa R} \right]_{\text{ret}} \\
&= \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{(\mathbf{v} \cdot \hat{\mathbf{R}}) \hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}} \\
&= \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{\kappa R} (1 - \kappa) \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}} \\
&= \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} - \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}},
\end{aligned}$$

and

$$\begin{aligned}
\frac{[R]_{\text{ret}}}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} &= \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R} \right]_{\text{ret}} - \frac{1}{c} \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} \frac{\partial}{\partial t} [R]_{\text{ret}} \\
&= \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R} \right]_{\text{ret}} + \frac{\mathbf{v} \cdot \hat{\mathbf{R}}}{\kappa c} \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}},
\end{aligned}$$

the Feynman expression now becomes

$$\begin{aligned}
\mathbf{E}(\mathbf{x}, t) &= \frac{q}{4\pi\epsilon_0} \left\{ \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} + \frac{[R]_{\text{ret}}}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} + \frac{1}{c^2} \frac{\partial^2}{\partial t^2} [\hat{\mathbf{R}}]_{\text{ret}} \right\} \\
&= \frac{q}{4\pi\epsilon_0} \left\{ \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R} \right]_{\text{ret}} + \frac{\mathbf{v} \cdot \hat{\mathbf{R}}}{\kappa c} \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} \right. \\
&\quad \left. + \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} - \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}} \right\} \\
&= \frac{q}{4\pi\epsilon_0} \left\{ \left[ \frac{\hat{\mathbf{R}}}{R^2} \right]_{\text{ret}} \left[ 1 + \frac{\mathbf{v} \cdot \hat{\mathbf{R}}}{\kappa c} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}} \right\} \\
&= \frac{q}{4\pi\epsilon_0} \left\{ \left[ \frac{\hat{\mathbf{R}}}{\kappa R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[ \frac{\hat{\mathbf{R}}}{\kappa R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[ \frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}} \right\},
\end{aligned}$$

which is the Jefimenko expression.